



Hospital Engineering and Management

Health Care Hierarchy in India and A Case Study on Air Borne Hospital

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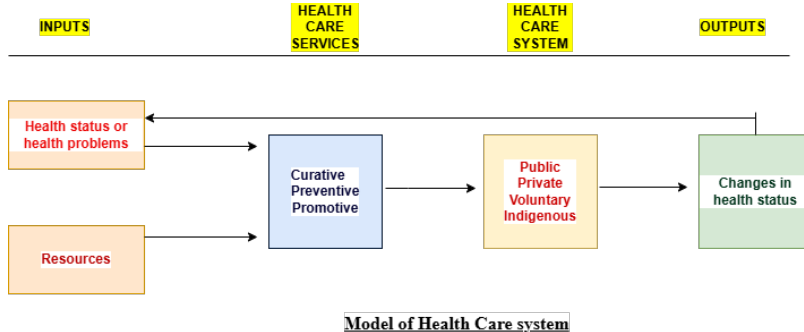
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Indian Hierarchy of Hospitals and Health Centres

Source: K. Park, *Textbook of Preventive and Social Medicine*



Indian Health System



Overview: The Three-Tier Rural Health System

India's rural health infrastructure is organized in three tiers below the district level:

The Three Tiers

1. Sub-Centre (SC) – most peripheral;
2. Primary Health Centre (PHC);
3. Community Health Centre (CHC).

Above These

Sub-District Hospital → District Hospital → State / National level institutions.

Sub-Centre (SC)

The most peripheral contact point between the community and the public health system.

Population Norm

- ▶ 1 per 5,000 (plains);
- ▶ 1 per 3,000 (hilly, tribal, desert areas).

Staffing

ANM / Health Worker (Female), Health Worker (Male); ASHA at the village/habitation level.

Services

Maternal and child health, immunization, family planning, basic curative care, health education.

Primary Health Centre (PHC)

The first rural institution staffed by a qualified Medical Officer.

Population Norm

- ▶ 1 per 30,000 (plains);
- ▶ 1 per 20,000 (hilly/tribal/difficult areas).

Facilities

4–6 beds, basic diagnostics, OPD, essential drugs.

Referral Role

Acts as a referral unit for about 6 Sub-Centres; refers cases onward to CHC and higher-order hospitals.

Community Health Centre (CHC)

A block-level referral hospital positioned above the PHC – the First Referral Unit (FRU).

Population Norm

- ▶ 1 per 1,20,000 (plains);
- ▶ 1 per 80,000 (hilly/tribal areas).

Capacity

30 beds; four specialists – Surgeon, Physician, Obstetrician-Gynaecologist, Paediatrician.

Facilities and Role

Operation theatre, X-ray, labor room, blood storage, emergency services; serves as FRU for about 4 PHCs.

Sub-Divisional and District Hospitals

Sub-Divisional Hospital

Serves a group of blocks; provides intermediate-level specialist care.

District Hospital

- ▶ apex referral institution for the district;
- ▶ wider range of specialties, higher bed strength;
- ▶ headed by the Civil Surgeon / Chief Medical Officer;
- ▶ empanelled under schemes such as PM-JAY (Ayushman Bharat) for cashless treatment.

Ayushman Arogya Mandirs (Health and Wellness Centres)

Sub-Centres and PHCs are being progressively upgraded under Ayushman Bharat.

Renaming

Earlier known as Health and Wellness Centres – now Ayushman Arogya Mandirs.

Expanded Service Package

NCD screening, mental health, elderly care, free diagnostics and medicines.

Punjab Health Care Facilities: Sub-Centre and PHC



Sub-Centre (SC)



Primary Health Centre (PHC)

Punjab Health Care Facilities: CHC and Tertiary Care Hospital

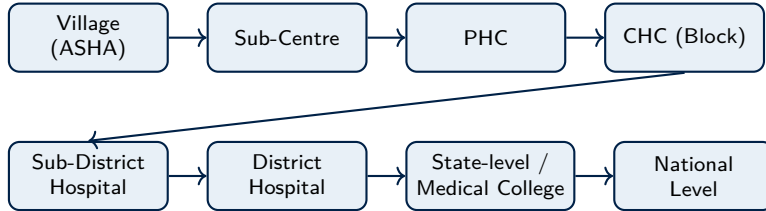


Community Health Centre (CHC)



Tertiary Care Hospital

Summary Diagram: The Complete Hierarchy



Referral flow, resources, specialization and population coverage increase up the hierarchy.



Airborne Hospital

60 Para Field Hospital

A Case Study in Rapid-Deployment Military Medicine



Origin and History

Raising and Early Service

Raised on 10 August 1942 as **60 Indian Field Ambulance** at Secunderabad; saw action in Burma with 2 Indian Airborne Division, which led to its selection for the parachute role.

The Korean War (1950–54)

- ▶ deployed as **60 Parachute Field Ambulance** under Lt Col A. G. Rangaraj, as India's contribution to the UN Command;
- ▶ landed at Pusan on 20 November 1950; served for over three years – the longest single tenure of any unit in the war;
- ▶ treated over 200,000 casualties, including 2,300 field surgeries; affectionately known as the “angels in maroon berets.”

Recognition

Awarded the President's Trophy by Dr. Rajendra Prasad on 10 March 1955 – the only unit ever to receive it.

Role and Operating Concept

Core Identity

- ▶ a parachute-trained field hospital of the Army Medical Corps;
- ▶ medics who are also qualified paratroopers, able to jump with combat forces;
- ▶ held at very high readiness – “anytime, anywhere.”

Deployment Philosophy

- ▶ can deploy as a full field hospital, or in modular components depending on requirement;
- ▶ responds to both national (internal) and international (HADR) emergencies;
- ▶ works alongside NDRF, IAF and Indian Navy assets during disaster response.

Structure

Team Composition (typical mission)

Surgeons, anaesthesiologists, orthopaedic surgeons, dental and general physicians, paramedical staff, and logistics / communication support – team size scales from roughly 40 to over 100 personnel depending on the mission.



Source: @adjpi, Indian Army

Services Provided

Clinical Services

Trauma and emergency surgery, orthopaedic and general surgery, dental care, women and child care, and public health support (water purification, communications).



Source:@adjpi, Indian Army



Source: @adjpi, Indian Army

BHISHM Cube: World's First Portable Hospital



Source: @adjpi, Indian Army

What It Is

BHISHM – Bharat Health Initiative for Sahyog, Hita & Maitri – is the world's first indigenous, modular, portable emergency hospital.

Under Project Aarogya Maitri

Jointly developed by the Ministry of Health, Ministry of Defence and the National Security Council; can be deployed within minutes to treat up to 200 casualties.

Operation Dost – Turkey–Syria (2023)

Trigger

A devastating 7.8-magnitude earthquake struck Turkey and Syria on 6 February 2023.

- ▶ a field hospital was established at Iskenderun in Hatay Province;
- ▶ the facility became operational within days of the disaster;
- ▶ the team provided sustained trauma care and surgery for survivors;
- ▶ operations concluded with formal appreciation from the local population.

60 PARA Mob



Source: @adjpi, Indian Army

Operation Brahma – Myanmar (2025)

Trigger

A 7.7-magnitude earthquake struck Myanmar and Thailand on 28 March 2025, killing over 1,600 people.

- ▶ 118-member Field Hospital Unit (“Airborne Angels Task Force”) airlifted to Mandalay by IAF C-17 aircraft;
- ▶ deployed with over 60 tonnes of supplies and rapidly-erectable surgical shelters;
- ▶ established a large Medical Treatment Centre offering trauma, surgical and women-and-child-care services;
- ▶ one of the first international responders on the ground.

Operation Sagar Bandhu – Sri Lanka (2025)

Trigger

Cyclone Ditwah caused severe flooding and landslides in Sri Lanka in late November 2025.

- ▶ full-fledged Para Field Hospital airlifted to Mahiyanganaya, Uva Province (2–12 December 2025);
- ▶ 78–85 member medical and logistics team;
- ▶ treated over 7,000 civilians – medical, surgical and dental care;
- ▶ also restored local communication and water- purification infrastructure.

Recognition

Formally commended by the Sri Lanka Army at Panagoda Cantonment on 14 December 2025.

Operation Amistad – Venezuela (2026)

Trigger

Twin earthquakes (7.2 and 7.5 magnitude) struck Venezuela on 24 June 2026, causing extensive loss of life.

- ▶ 41-member contingent flown in on two IAF C-17 Globemaster III aircraft (26 June 2026);
- ▶ carried two BHISHM Cubes, 6 tonnes of medicines and 30 tonnes of relief supplies;
- ▶ conducted over 8,000 medical procedures, including 20+ major surgeries;
- ▶ mission concluded successfully in July 2026, with the Venezuelan government's formal thanks.

60 Para Field Hospital in Action



Source: @adjpi, Indian Army

Recognition and Achievements

Historic Honours

- ▶ President's Trophy (1955) – the only unit to receive it;
- ▶ Maha Vir Chakra to Lt Col A. G. Rangaraj;
- ▶ 25 Mentioned-in-Dispatches; US Bronze Star and UN citations.

Recent Global Recognition

DG Armed Forces Medical Services has highlighted the unit's "anytime, anywhere" readiness and its consistent humanitarian impact across Turkey, Myanmar, Sri Lanka and Venezuela.

Recap: Key Takeaways

- ▶ India's tiered public health structure ensures graded referral from village to district level;
- ▶ military medicine adds a rapid-deployment layer – the 60 Para Field Hospital can establish a fully functional hospital within days, anywhere in the world;
- ▶ modular systems such as BHISHM Cubes are reshaping how mass-casualty and disaster medicine is delivered;
- ▶ both systems – civilian and military – reflect the same engineering challenge: reliable, safe care delivered under constrained conditions.

Thank You

Questions and Discussion



Assignment: Disaster Management Planning in a Hospital

5 Marks

Task

Develop a disaster-management plan for a hypothetical 100-bed rural hospital near Patiala. Available specialties include medicine, surgery, anaesthesia, gynaecology, paediatrics, laboratory services, X-ray and ultrasound.

Scenario and Planning Requirements

1. A rail accident occurs 5 km from the hospital with casualties flowing in to your hospital
2. Define the triage or casualty priority strategy;
3. Command structure – who is in charge, and of what, during the emergency;
4. Estimate bed and staff surge requirements, including expansion beds, staff recall and resource allocation;
5. Provide an evacuation plan, if required.

Submission

Submit a 2–3 page plan. A flowchart or block diagram is encouraged.